

Solution Requirements

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Name	AI Study Buddy API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Students can register and securely log in to access the system.</i>	High
2	Authorization levels	<i>Students can access study features and their own learning data.</i>	Medium
3	External interfaces	<i>The system connects the frontend, backend, AI processing module, and database.</i>	High

4	Transactions processing	<i>The system receives student questions, processes them using AI, and displays responses.</i>	High
5	Reporting	<i>The system provides quizzes, flashcards, summaries, and study plans for learning support.</i>	High
6	Business rules, etc.	<i>The AI should provide relevant and easy-to-understand explanations for student queries.</i>	High
7	Compliance to laws or regulations	<i>Protect student information and maintain data privacy.</i>	Medium
8	<i>Other</i>	<i>Students can upload study materials and use them for learning assistance.</i>	Medium

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should respond quickly to student queries.</i>	<i>Response within 3 seconds under normal load</i>

2	Scalability	<i>The system should support increasing numbers of students and study data.</i>	Support 100+ concurrent users
3	Security & Data Privacy	<i>User information and study data should be protected.</i>	<i>Secure login and protected database access</i>
4	Reliability & Availability	<i>The system should remain available and handle errors properly.</i>	99% availability <i>during normal operation</i>
5	Usability & Accessibility	<i>The interface should be simple, clear, and easy for students to use</i>	<i>User can access major features with minimal steps</i>
6	<i>Other</i>	<i>The system should be easy to update and maintain.</i>	Modular frontend, backend, AI, and database components